Statistics – 7

Probability Distributions

12 April 2026 · Lecture Notes · Statistics-7

Bernoulli	Binomial	Poisson	Exponential	Uniform

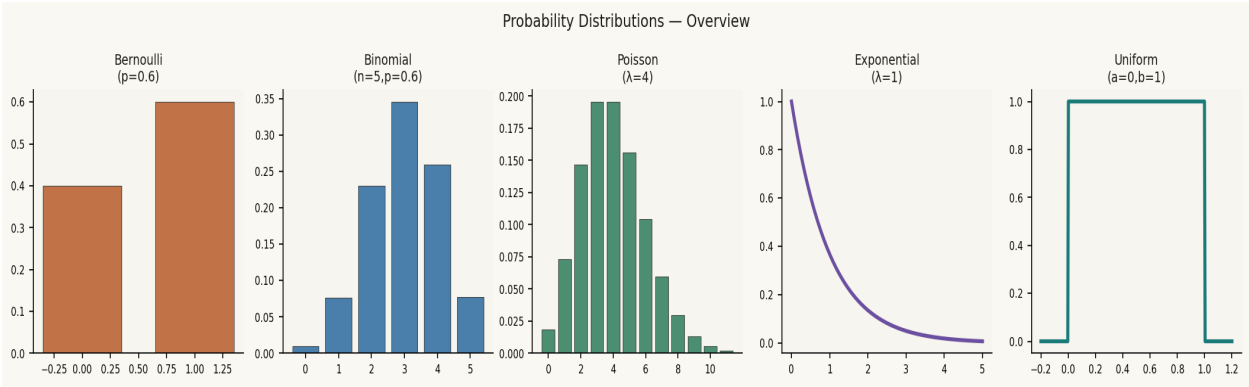

Figure 1: Overview of all 5 probability distributions covered in this class

Distribution	Type	Models	Parameter(s)	Formula
Bernoulli	Discrete	Single trial, 2 outcomes	p	P(X=x) = p^x (1-p)^(1-x)
Binomial	Discrete	n Bernoulli trials	n, p	P(X=k) = C(n,k) p^k (1-p)^(n-k)
Poisson	Discrete	Count of events in interval	λ (rate)	P(X=k) = e^(-λ) λ^k / k!
Exponential	Continuous	Time between events	λ (rate)	P(T≤t) = 1 – e^(–λt)
Uniform	Continuous	Equal likelihood in range	a, b	f(x) = 1/(b-a) for x∈[a,b]

§ 01 Bernoulli Distribution

DEFINITION

The **simplest possible distribution** — a single trial with exactly **two outcomes**.

→ 1 = Success (with probability p)

→ 0 = Failure (with probability 1–p) [Note: 0 does NOT mean failure in a negative sense]

Only one parameter: **p** (the probability of success).

Examples: every coin flip → yes/no | every question → pass/fail | job application → get/don't get

PMF Formula

```
P(X = x) = p^x · (1–p)^(1–x) x ∈ {0, 1}
x = 0 → p■ · (1–p)¹ = 1·(1–p) = 1–p
x = 1 → p¹ · (1–p)■ = p·1 = p
```

Mean and Variance:

```
Mean = p Variance = p(1–p)
```

Worked Example — 60% Customer Shoe Purchase

On average, 60% of customers buy shoes. p = 0.6

Outcome	X value	Probability	Meaning
Success (buys shoe)	1	P(X=1) = 0.6	Mean = p = 0.6
Failure (no buy)	0	P(X=0) = 0.4	Variance = 0.6×0.4 = 0.24

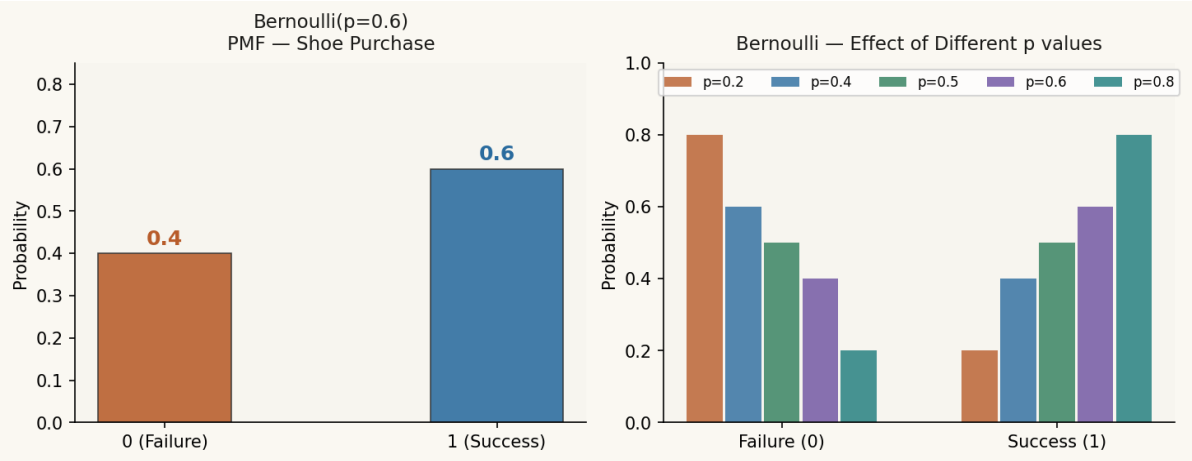

Figure 2: Bernoulli PMF (p=0.6) and comparison across different p values

§ 02 Binomial Distribution

DEFINITION

Bernoulli is discrete with only **two values** (1 trial).

Binomial models n Bernoulli trials — it's n independent Bernoulli experiments combined.

Binomial(n, p): n = number of trials, p = success probability per trial.

→ Coin toss (X → head): 1 toss = Bernoulli. N tosses = Binomial.

PMF Formula

```
P(X = k) = C(n,k) · p^k · (1–p)^(n–k)
C(n,k) = n! / [(n–k)! · k!] (n choose k = number of arrangements)
k = exact successes wanted | n = total trials | p = success prob
```

Why the Combination C(n,k)?

INTUITION

3 buyers out of 5 customers can happen in many orders:

BBBNN | BNBBN | NBBBN | ... etc.

C(5,3) = 10 counts ALL such arrangements.

Combination × P(one specific arrangement) = total probability.

Worked Example — Pizza Brand Garlic Dip

Each customer buys garlic dip with p = 0.6. Today 5 customers arrive. What is the prob that **exactly 3** customers buy?

Part	Calculation	Value
Step 1: Combination	C(5,3) = 5! / (2! × 3!) = 120/12	= 10
Step 2: Success part	$p^3 = 0.6^3$	= 0.216
Step 3: Failure part	(1–p)^(5–3) = 0.4^2	= 0.16
Step 4: Combine all	P(X=3) = 10 × 0.216 × 0.16 = 10 × 0.03456	≈ 0.3456

$$P(X = 3) = C(5,3) \times 0.6^3 \times 0.4^2 = 10 \times 0.216 \times 0.16 \approx 0.3456$$

Assignment: P(X ≥ 3) — At least 3 buyers

ASSIGNMENT SOLUTION

$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$

$P(X=3) \approx 0.3456$ (calculated above)

$P(X=4) = C(5,4) \times 0.6^{\blacksquare} \times 0.4^{1} = 5 \times 0.1296 \times 0.4 \approx 0.2592$

$P(X=5) = C(5,5) \times 0.6^{\blacksquare} \times 0.4^{\blacksquare} = 1 \times 0.07776 \times 1 \approx 0.0778$

$P(X \geq 3) = 0.3456 + 0.2592 + 0.0778 \approx 0.6826 \rightarrow$ ~68.3% chance

Alternative: $P(X \geq 3) = 1 - P(X \leq 2) = 1 - [P(0)+P(1)+P(2)]$

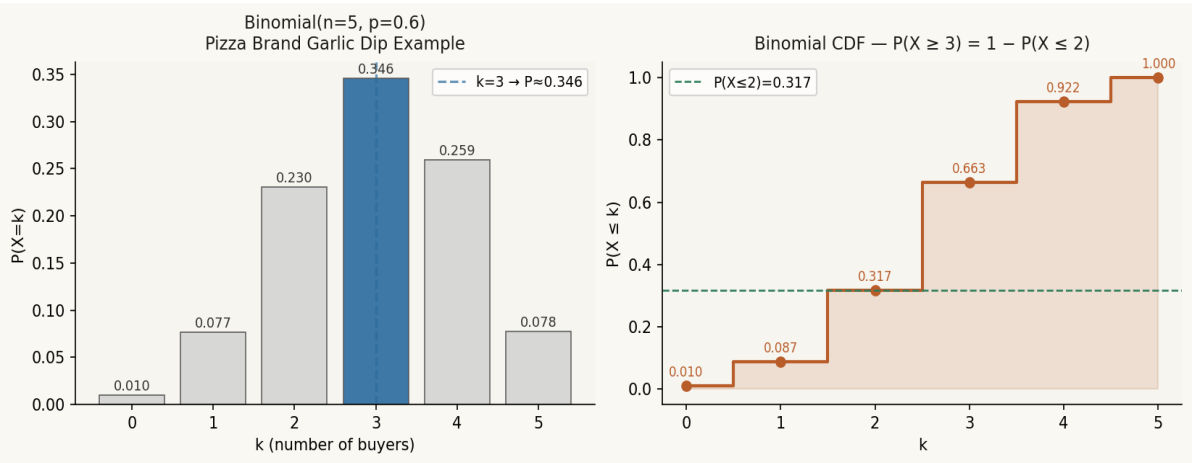

Figure 3: Binomial(n=5, p=0.6) PMF and CDF — P(X=3)≈0.346, P(X≥3)≈0.683

§ 03 Poisson Distribution

DEFINITION

Models the **count of rare events occurring at a constant rate (λ)** in a fixed interval.

Discrete: values are non-negative integers (0, 1, 2, ...)

Parameter: λ = average rate of events (mean)

Memoryless: events occur independently.

Examples: web traffic → visits per minute | customer service → calls per hour

Note: Poisson → ML → robust (used in ML models)

PMF Formula

```
P(X = k) = e^(−λ) · λ^k / k!
λ = average rate (mean) | k = number of events | e = 2.7182...
```

Worked Example — Website Visits

A website averages **4 visits per hour**. λ = 4. What is P(exactly 6 visits in an hour)?

```
k = 6, λ = 4
P(k=6) = e^(−4) · 4^6 / 6! = e^(−4) · 4096 / 720 ≈ 0.104
→ ~10.4% chance of getting exactly 6 visits
```

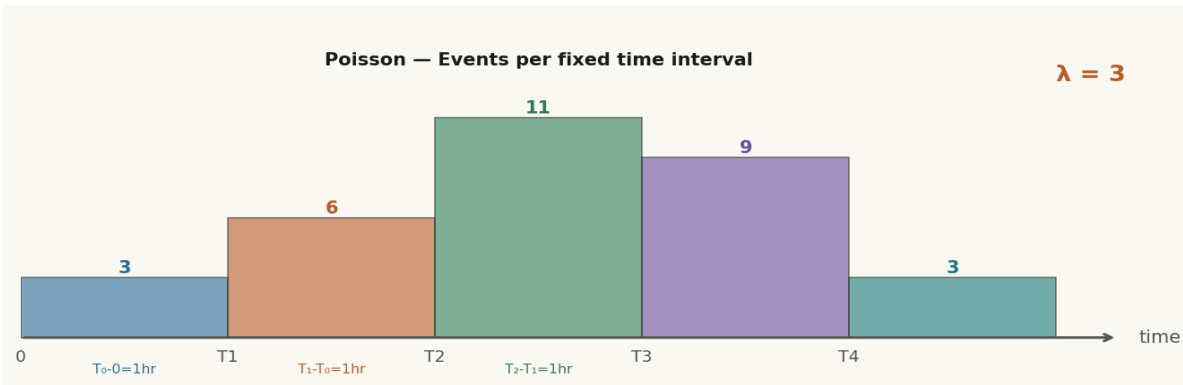

Figure 4: Poisson — events counted per fixed time interval (λ=3 events/interval)

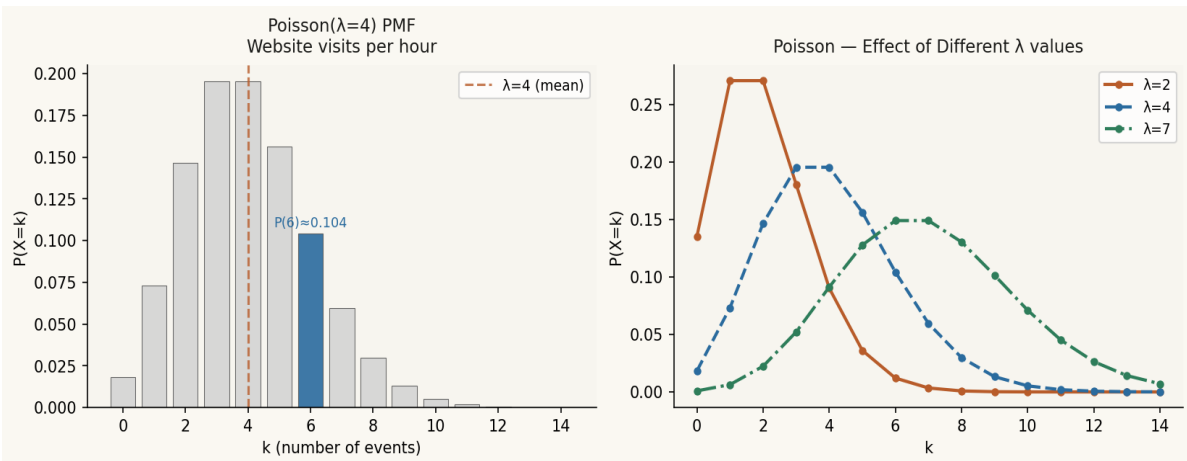

Figure 5: Poisson(λ=4) PMF and comparison across λ values

§ 04 Exponential Distribution

DEFINITION

Models the **time between events in a Poisson process**.

Poisson ↔ Exponential are paired: Poisson=count, Exponential=time gap.

Poisson: How many events happen in a fixed time?

Exponential: How much time until the next event?

Continuous: any positive real value.

Examples: call centre → time between two calls | web server → time between two user logins

CDF Formula

```
P(T ≤ t) = 1 − e^(−λt)
T = waiting time until next event | λ = rate (from Poisson) | t = time threshold
```

Worked Example 1 — Server Requests

On average, **3 requests per minute**. What is P(next request comes within 10 seconds)?

Step	Calculation	Value
λ (Poisson rate)	3 req/minute	3/min
λ per second	3/60	= 0.05/sec
t	10 seconds	= 10
P(T ≤ 10)	1 – e^(–0.05 × 10) = 1 – e^(–0.5)	≈ 0.393 (39%)

```
P(T ≤ 10) = 1 − e^(−0.5) ≈ 0.393 → ~39% chance of request within 10 seconds
```

Worked Example 2 — Server Crash

A server crashes on average **0.1 times per day**. What is P(runs > 10 days without crashing)?

Step	Calculation	Value
λ	0.1 crashes/day	= 0.1
t	10 days	= 10
P(T ≤ 10) = P(crash)	1 – e^(–0.1 × 10) = 1 – e^(–1)	≈ 0.6321
P(T > 10) = P(no crash)	1 – P(T ≤ 10) = 1 – 0.6321	≈ 0.3679 (37%)

P(T > 10) = 1 – P(T ≤ 10) = e^(–λt) = e^(–1) ≈ 0.3679 → ~37% chance of no crash

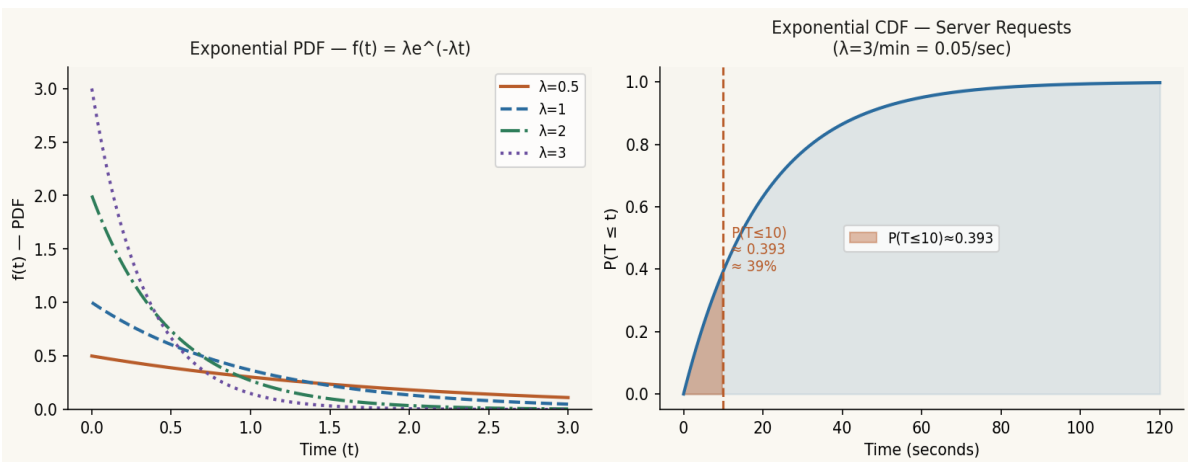

Figure 6: Exponential PDF (multiple λ) and CDF for server request example

§ 05 Uniform Distribution

DEFINITION

Every value in an interval [a, b] is **equally likely**.

The PDF is a **flat rectangle** with constant height = 1/(b–a).

Area of rectangle = height × width = 1/(b–a) × (b–a) = **1** (total prob = 1).

Examples: rolling a dice {1,2,3,4,5,6} | bus arrives anywhere between 10:00–10:30 | random arrival time between 1 PM–2 PM

PDF Formula

```
f(x) = 1/(b–a) for x ∈ [a, b]
height = 1/(b–a) | width = b–a | height × width = 1 (area = total prob)
P(x■ ≤ X ≤ x■) = (x■ – x■) × height = (x■ – x■) / (b–a)
```

Worked Example — Delivery Arrival Time

A rider can arrive anytime between 20–30 minutes. X ~ Uniform(20, 30)

Q: What is P(delivery arrives between 22 and 25 minutes)?

Step	Calculation	Value
a, b	a = 20, b = 30	Range = 10
PDF height	1/(b–a) = 1/10	= 0.1
Width of interest	25 – 22	= 3
P(22 ≤ X ≤ 25)	width × height = 3 × 0.1	= 0.3 (30%)

```
P(22 ≤ X ≤ 25) = area of rectangle = 3 × 0.1 = 0.3 → 30% chance
```

Discrete Uniform — Dice Roll

Dice = {1,2,3,4,5,6} → each face has density/probability = 1 (equally likely).

Bus arrives anywhere between 10:00 and 10:30 → every 5-minute slot equally likely. PDF is flat rectangle from 10:00 to 10:30.

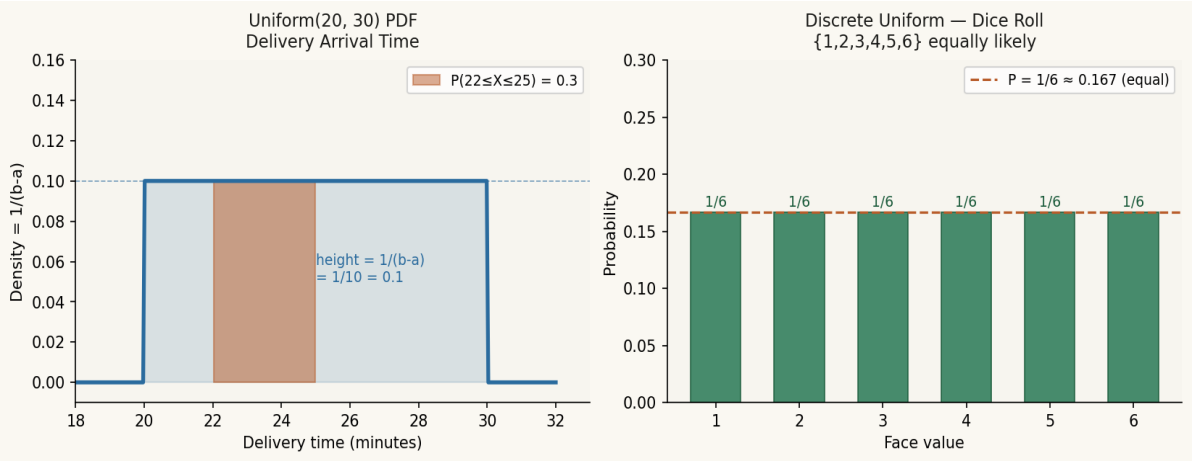

Figure 7: Uniform PDF for delivery time (left) and discrete uniform dice roll (right)

§ 06 Summary — All Distributions

Distribution	Type	Key Formula	Mean	Variance	Real Example
Bernoulli(p)	Discrete	P(X=x)=p^x(1-p)^(1-x)	p	p(1-p)	Job offer: yes/no
Binomial(n,p)	Discrete	C(n,k)·p^k·(1-p)^(n-k)	np	np(1-p)	3 of 5 buy garlic dip
Poisson(λ)	Discrete	e^(-λ)·λ^k / k!	λ	λ	6 web visits/hour
Exponential(λ)	Continuous	P(T≤t)=1–e^(-λt)	1/λ	$1/\lambda^2$	Next req in 10 sec
Uniform(a,b)	Continuous	f(x)=1/(b-a)	(a+b)/2	$(b-a)^2/12$	Delivery 22-25 min

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